DATA ITEM DESCRIPTION

Title: DATA REQUIREMENTS FOR SPACE DEBRIS MITIGATION ASSESSMENTS

Number: DI-SAFT-82416
AMSC Number: F10431
DTIC Applicable: No
Preparing Activity: 19
Applicable Forms: N/A

Approval Date: 20231004
Limitation: N/A
GIDEP Applicable: No
Project Number: SAFT-2023-002

Use/Relationship:
Data Requirements for Space Debris Mitigation Assessments to support the development of a Launch Vehicle (LV)/Launch Service Space Debris Assessment Report (SDAR) and/or Space Vehicle (SV) Space Debris Assessment Report/End-of-Life Plan (SDAR/EOLP), IAW DAFI 91-202, *The US Air Force Mishap Prevention Program* including implementation changes per DAFI91-202_DAFGM2023-01, Department of the Air Force Guidance Memorandum dated 13 April 2023. Data will be used by the SSC AATS/LV or by the SSC/SV Program Office System Safety Managers (SSMs) to assess if contractors' United States (U.S.) Space Force LV or SV orbital system and mission design follows *National Space Policy* (NSP) and, U.S. Government (USG) *Orbital Debris Mitigation Standard Practice* (ODMSP). (The most current version of ODMSP and DAFI 91-202 in force can be obtained at: https://orbitaldebris.jsc.nasa.gov/mitigation/ and https://www.e-publishing.af.mil/, respectively.)

This Data Item Description (DID) contain the format, content, and intended use of the information for the data the contractor will need to develop and deliver to support assessment as defined by the contract statement of work (SOW).

In this document, "SV[1]" refers to space vehicles, and "LV" refers to launch vehicle stage(s) that become orbital. Both SVs and LV stages are addressed. "Orbital stage components" includes items released from the orbital stage, including adapters, spacer rings, etc.

Requirements:

1. **Reference Documents.** The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.
2. **Format.** Data Requirements for Space Debris Mitigation Assessments shall be in contractor's format.
3. **Content.** Data Requirements for Space Debris Mitigation Assessments shall contain the following:

   3.1. A listing of all objects with longest dimension greater than 5 millimeters (mm) planned to be released into Earth orbit during normal operations of the SV or LV stage, along with the following information about each object:

[1] Synonymous with spacecraft (SC) when appears in any related documentation or reference

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3.1.1. Rationale and a description of the necessity for release.
3.1.2. For an LV stage, time of release from launch for LV stage.
3.1.3. For an SV, time of release from deployment of the SV from an LV stage.
3.1.4. Release velocity with respect to the SV or LV stage.
3.1.5. Expected orbital parameters (apogee, perigee, and inclination) after release from the LV stage or SV.
3.1.6. Calculated orbital lifetime until reentry into Earth's atmosphere.
3.1.7. Material composition, dimensions, and mass of object.

3.2. Total SV mass at the start of the mission, at end-of-life prior to disposal, and at any events that separate distinct phases of the mission.

3.3. Failure modes and effects analyses (FMEA) results, demonstrating that the integrated probability of explosion for all failure modes (excluding collisions) of each separate SV and orbital stage is < 0.001.

3.4. Passivation methods and processes
3.4.1. List of internal energy sources for the SV or orbital stage that could cause an explosion (e.g., pressurized tanks, extra fuel or oxidizer, batteries, momentum wheels).
3.4.2. Description of the plan for eliminating these internal energy sources as part of the end-of-life disposal plan.

3.5. Expected timelines
3.5.1. For the LV, in appropriate units, from launch time (T-0), of the following events:
3.5.1.1. Stage separation from the LV.
3.5.1.2. SV separation from the LV stage.
3.5.2. For the SV orbit transfer and flight profile plan, in appropriate units, from LV separation time (T-0), of the following events:
3.5.2.1. SV mission events that cause changes in the orbital parameters (needed for assessment of Probability of Collision With Large Objects [PCWLO]).
3.5.2.2. End-of-Life disposal.
3.5.2.3. Deployment of tether or atmospheric drag enhancement device (if equipped).
3.6. The launch trajectory/flight profile plan
3.6.1. Mission orbit(s) parameters (orbital elements) and time durations spent in each mission orbit until initiation of disposal action.
3.6.2. Disposal orbit(s) parameters (orbital elements) and time durations spent in each distinct phase of the disposal profile until final disposal orbit is achieved.
3.6.3. Planned trajectory file, if orbital elements or mass vary significantly due to frequent or constant maneuvering (e.g., during orbit transfer).
3.6.4. Description of orbit maintenance or station-keeping planned for the mission duration. Describe how the orbit is going to be maintained via maneuvers or other means (e.g., electrodynamic tethers, solar sailing, differential drag control) during each phase of the mission.
3.6.5. Description of any collision avoidance process that is going to be performed during orbit transfer or mission operation, including controlled disposal (reentry or on-orbit). This description includes the criteria used to determine whether a collision avoidance action is performed, e.g., threshold on probability of collision at conjunction, miss

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distance, whether orbit transfer maneuver uncertainty is going to be accounted for (e.g., via covariance or Monte Carlo modeling).

3.7. The SV and orbital stage components data (bill of materials), including:
  3.7.1. Physical dimensions (i.e., length, height, width, mass) of all dominant SV or LV stage components, to include: the SV bus or LV stage body, solar arrays, antennas, gravity gradient booms, adapter rings, etc.
  3.7.2. For items that have many gaps (e.g., trusses), effective projected area of solid portions or an accurate description of the item construction (e.g., a Computer-Aided Design (CAD)finite element model in a standard format)

3.8. Data on SV components exposed to Micro-Meteoroids and Orbital Debris (MMOD) (not needed for LV stages):
  3.8.1. Tabulated list of components and surfaces exposed to MMOD that are needed to execute end-of-life disposal functionality (e.g., maneuvers, tether or sail deployment, propellant depletion), including components mounted to the inside of panels exposed to MMOD (since MMOD could penetrate through the panel and component wall).
  3.8.2. Description of redundancy of these components.
  3.8.3. Data needed for each of these components to assess risk of damage by MMOD, including:
    3.8.3.1. Criteria for MMOD to cause failure of the component (surface penetration depth, penetration hole diameter, number of penetrations or holes, probability of failure given a penetration has occurred, etc.)
    3.8.3.2. Exposed area of each component and surface.
    3.8.3.3. Estimate of the fraction of the surface area of each component and surface that is not blocked if an item is partially shielded from MMOD (e.g., less than 2*pi steradian field of view (FOV) for a flat surface).
    3.8.3.4. For each single exposed surface for which penetration could prevent disposal: thickness and material (e.g., aluminum).
    3.8.3.5. For each single exposed surface on which cratering could prevent disposal (e.g., optical surfaces on sun sensors or star trackers): impact crater size or area that would prevent adequate operation of the component housing the optical surface for post-mission disposal, and material (e.g., fused silica glass, fused quartz glass, polycarbonate).
    3.8.3.6. For each dual-wall or multi-wall configuration (e.g., outer wall is shielding an inner wall whose penetration could prevent disposal): thickness and material of both walls, and spacing distance between walls. (This is needed to model the multiwall shock effect.)
    3.8.3.7. For each exposed panel that is a honeycomb sheet: thickness and material of front and back panels and honeycomb panels connecting those two, and spacing distance between all panels.
    3.8.3.8. For each harness or cable that is exposed to MMOD, provide a description of the cable design. For example, provide the Cable Type as described in D. Jex, A. Adkinson, J. English, and C. Linebaugh, "Hypervelocity Impact Testing of Cables," NASA TN D-7178, February 1973.
    3.8.3.9. For each surface covered with multi-layer insulation (MLI) that is exposed to MMOD, provide the mass per unit area of MLI (or description with sufficient detail to compute this) and spacing distance of MLI from covered surface.

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3.8.3.10. For pressurized vessels (e.g., Composite Overwrapped Pressure Vessel (COPV)): Maximum penetration depth of vessel wall before wall rupture occurs due to pressurized fluid.

3.8.3.11. Surface orientation in the SV bus frame (i.e., surface normal vector).

3.8.4. Description of the SV bus attitude motion relative to a relevant standard frame (e.g., local vertical/local horizontal frame, Earth-centered inertial frame, Sun-centered inertial (ecliptic) frame. (This is needed so that the fraction of time spent by the surface in each orientation can be determined.)

3.9. Tabulated list of items needed to assure post mission disposal reliability, including:

3.9.1. Systems, subsystems, and components required to accomplish post mission disposal operations

3.9.2. End of mission reliability assessment results for each system, subsystem, and vehicle (SV and upper stage) component

3.9.3. Reliability assessments should account for failures occurring from start of mission to completion of disposal, including the entire mission design life.

3.10. Data for uncontrolled reentry hazards assessment, including:

3.10.1. Tabulated list of parts (e.g., a master equipment list – see Example Space Vehicle Data Template on page 6) for each reentering vehicle (SV and upper stage), to include:

3.10.1.1. Mass and quantity of each part

3.10.1.2. Material composition of each part (e.g., 2024 Aluminum, 410 Stainless Steel, Kevlar, graphite-epoxy, titanium alloy Ti-6Al-4V))

3.10.1.3. Physical dimensions for each part (including height, width, length, wall thickness; for honeycomb panels, also include face-sheet thicknesses, honeycomb density)

3.10.1.4. Illustrations or graphics of the shape of each part

3.10.1.5. Illustrations or graphics showing placement of the parts in relation to each other

3.10.1.6. Parent/Child relationship for each part

3.10.2. Classical orbital elements, or best estimates, at end-of-mission for each reentering vehicle

3.10.3. Expected year of reentry from an orbital lifetime analysis, assuming a nominal atmospheric profile, for each reentering vehicle

3.11. Data for controlled reentry assessment, including:

3.11.1. Description of vehicle (SV and upper stage), to include:

3.11.1.1. Mass of the vehicle both pre-and-post reentry maneuver

3.11.1.2. Propulsive capability of the vehicle (thrust and specific impulse)

3.11.1.3. Delta-V and propellant mass required to accomplish post-mission disposal

3.11.1.4. Physical dimensions of the vehicle (including length, width, height, or diameter)

3.11.1.5. Illustration or graphic of the vehicle

3.11.1.6. Expected debris casualty area (Ac) associated with reentry or breakup

3.11.2. Reentry trajectory profile with targeted impact location

3.11.3. Classical orbital elements, or best estimates, at end-of-mission for each reentering vehicle

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3.11.4. Assessment of reentry maneuver reliability based on vehicle flight history, to include:
3.11.4.1. List of the number of successes, failures, partial failures, or significant anomalies for all prior missions of the LV
3.11.4.2. List of previous controlled reentry attempts of the LV and associated outcomes

3.12. Data for drag enhancement devices (if equipped) other than tethers, for controlled reentry assessment, including:
3.12.1. Expected projected area after deployment and percent confidence level for this deployment for each device
3.12.2. Dimensions, shape, material, and mass of each device
3.12.3. Time of deployment of each device after mission start
3.12.4. Procedure for drag device deployment and control until reentry
3.12.5. Construction characteristics that make each drag enhancement device resilient to debris and meteoroid impacts (e.g., balloon rigidization, compartments), and the expected size of debris or meteoroid that could make it non-functional as a drag enhancement device
3.12.6. Expected change in ballistic coefficient/drag profile stemming from deployment of drag device
3.12.7. Classical orbital elements, or best estimates, at end-of-mission for each reentering device

3.13. Data for tethers (if equipped), including:
3.13.1. Expected tether length after deployment and percent confidence level for this deployment for each tether
3.13.2. Width (if flat tape) or diameter (if round) of each tether
3.13.3. Mass of each tether or device
3.13.4. Tether end mass of each tether
3.13.5. Time of deployment of each tether after mission start
3.13.6. Tether construction that makes it resilient to debris and meteoroid impacts (e.g., material, braiding, multiple strands), and expected size of debris or meteoroid that could sever the tether or make it non-functional as a drag enhancement device
3.13.7. Expected change in ballistic coefficient/drag profile stemming from deployment of tether Classical orbital elements, or best estimates, at end-of-mission for each reentering tether or device
3.13.8. Data on prior on-orbit performance for each tether
3.13.8.1. Data-based rationale for each tether on the effect(s) any prior on-orbit performance has had on the current device design
3.13.8.2. Description of whether and how the device was damaged or severed by debris or meteoroid impacts or by collision with another object
3.13.8.3. Description of how successful the device was (e.g., did it successfully achieve expected reduction in orbital lifetime?).
3.13.8.4. Description of orbit operations plan for an electrodynamic tether (if applicable)

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Example Space Vehicle Data Template

Object Description	Shape	Count	Material Type	Diam/Width (m)	Length (m)	Wall/Core Thickness (m)	Face Sheet Thickness (m)	Height (m)	Mass (kg)	Total Mass (kg)	Notes:
STRUCTURE											
Side Panel - Full	Plate	3	Al 2024 HC	1.000	2.000	0.024	0.001		14.24	42.7	
Side Panel - Cutout	Plate	1	Al 2024 HC	1.000	2.000	0.024	0.001		12.81	12.8	
Top/Bottom Panel	Plate	1	Al 2024 HC	1.000	1.000	0.024	0.001		7.12	7.1	
Mid Panel - Cutout	Plate	1	Al 2024 HC	1.000	1.000	0.024	0.001		5.70	5.7	
Bottom Panel - Cutout	Plate	1	Al 2024 HC	1.000	1.000	0.024	0.001		2.61	2.6	
Solar Array Boom	Cylinder	2	Al 2024	0.025	2.025	0.025			0.72	1.4	
POWER											
Battery Box	Box	1	Al 2024	0.467	0.617	0.008		0.317	28.66	28.7	
Battery Box - Inner Frame	Plate	3	Al 2024	0.296	0.450	0.008			3.12	9.4	
Battery Box - Bracket 1	Plate	2	Al 2024	0.025	0.317	0.001		0.025	0.05	0.1	
Battery Box - Bracket 2	Plate	2	Al 2024	0.025	0.467	0.001		0.025	0.07	0.1	
Batteries	Cylinder	12	Nickel	0.146	0.296	0.001			1.91	22.9	
Solar Panel	Plate	2	Al 2024 HC	1.000	2.000	0.024	0.001		14.24	28.5	
PROPULSION											
LH2 Tank	Sphere	1	Ti-6Al-4V	0.500		0.001			10.00	10.0	
LO2 Tank	Cylinder	1	Al 2024	0.898	1.000	0.003			38.90	38.9	
ATTITUDE CONTROL											
Magnetic Torquer	Cylinder	3	Iron	0.045	1.000	0.003			3.33	10.0	
Star Tracker	Cylinder	2	Al 2024	0.200	0.450	0.007			6.50	13.0	
RWL Housing	Cylinder	4	Al 6061-T6	0.400	0.100	0.004			4.36	17.4	
RWL Flywheel	Disk	4	Ti-6Al-4V	0.300		0.010			3.14	12.5	
RWL Shaft	Cylinder	4	SS A286	0.020	0.091	0.010			0.23	0.9	
Gyroscope	Box	1	Al 6061-T6	0.246	0.330	0.011		0.176	9.90	9.9	
TELECOMMUNICATION											
X-Band Antenna	Disk	1	Al 2024	0.600		0.003			1.98	2.0	
X-Band Boom	Cylinder	1	CFRP	0.090	1.700	0.004			3.19	3.2	
S-Band Transponder	Box	2	Al 2024	0.100	0.200	0.015		0.100	3.27	6.5	
COMMAND AND DATA											
Computer	Box	1	Al 2024	0.300	0.600	0.001		0.200	2.01	2.0	
Data Storage	Box	1	Al 6061-T6	0.200	0.250	0.030		0.200	16.99	17.0	
Command	Box	1	Al 6061-T6	0.250	0.400	0.030		0.150	24.87	24.9	
Telemetry	Box	1	Al 2024	0.200	0.300	0.013		0.200	10.38	10.4	
THERMAL											
Louver - Blade	Plate	6	Al 2024	0.050	0.500	0.001			0.07	0.4	
Louver - Frame 1	Plate	2	Al 2024	0.020	0.340	0.003			0.06	0.1	
Louver - Frame 2	Plate	2	Al 2024	0.020	0.500	0.003			0.08	0.2	
Cold Plate	Plate	1	Al 2024	0.340	0.540	0.025			13.07	13.1	
Space Radiator	Plate	1	Al 2024	0.300	0.500	0.010			4.21	4.2	
Totals		71								358.7	

End of DI-SAFT-82416**.**

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